

Silent Failure of Adaptive Loss Balancing in Physics-Informed Inverse Problems: Diagnosis and a Fix

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ABSTRACT

Adaptive loss balancing is standard practice in physics-informed neural networks (PINNs), yet the weights it produces are only meaningful for losses that the balancing anchor can actually see. We show that when a loss term is decoupled from the anchor, gradient-norm balancers fail *silently*, and that the direction of the failure is fixed by the balancer's functional form: relative-norm balancers drive the weight to zero, switching the loss off, while inverse-ratio balancers (e.g. learning-rate annealing) divide by a vanishing gradient and drive the weight to infinity. Neither failure raises an error or perturbs the reported training loss. We formalise the mechanism, introduce a cheap diagnostic — the anchor-coupling ratio $c_k = \|\nabla_\phi \mathcal{L}_k\| / \|\nabla_\theta \mathcal{L}_k\|$ — and propose an anchor-robust wrapper that detects decoupled losses at run time and neutralises their weight without any manual intervention or prior knowledge of which loss is affected. On a 2D shallow-water inverse problem, the failures are severe: the same gradient-norm balancer yields a drainage-coefficient error of 52.9% when the prior decoupled loss is discarded, and a weight of 3.3×10^7 when it is over-weighted. The wrapper repairs both directions, reducing the parameter error from 52.9% to 0.030% and bounding the exploded weight at 1.0. We further show that in this weakly identified problem the recovered parameters are prior-determined rather than data-determined, so which balancer is used decides the answer — a caveat that applies whenever a PINN inverse problem is only weakly identified.

1. Introduction

Physics-informed neural networks (PINNs) (Raissi, Perdikaris and Karniadakis, 2019; Sirignano and Spiliopoulos, 2018) train a network by minimising a composite objective whose terms — PDE residual, data misfit, boundary and initial conditions, and, in inverse problems, parameter priors — have different scales and converge at different rates. How to weight these terms is a long-standing practical problem, and one that has attracted considerable attention as physics-informed learning has matured (Karniadakis, Kevrekidis, Lu, Perdikaris, Wang and Yang, 2021; Cuomo, Di Cola, Giampaolo, Rozza, Raissi and Piccialli, 2022; Blechschmidt and Ernst, 2021). A large family of methods balances the terms *adaptively*, using gradient statistics: learning-rate annealing (Wang, Teng and Perdikaris, 2021) sets each weight to the ratio of the largest gradient norm to the loss's own gradient norm; GradNorm (Chen, Badrinarayanan, Lee and Rabinovich, 2018) equalises gradient magnitudes across tasks; PCGrad (Yu, Kumar, Gupta, Levine, Hausman and Finn, 2020) and CAGrad (Liu, Liu, Jin, Stone and Liu, 2021) resolve gradient conflicts. A second family avoids gradients entirely and balances using the loss values themselves, of which ReLoBRaLo (Bischof and Kraus, 2025), SoftAdapt (Heydari, Thompson and Mehmood, 2019) and uncertainty weighting (Kendall, Gal and Cipolla, 2018) are representative. These techniques are shared with the wider multi-task learning literature (Ruder, 2017; Crawshaw, 2020).

All gradient-based balancers share a design decision that is rarely discussed: the *anchor*, i.e. the parameter subset with respect to which gradient norms are measured. Using the full parameter vector is expensive — one backward pass per loss term — so implementations routinely restrict the anchor to a subset, most commonly the last layer.

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This paper is about what happens when that shortcut meets a loss the anchor cannot see. Consider an inverse problem with a parameter prior $\mathcal{L}_{\text{prior}} = \sum_j w_j (\mu_j - \mu_j^*)^2$ on physical parameters μ . If the anchor is the network’s last-layer weight matrix W_L , then $\nabla_{W_L} \mathcal{L}_{\text{prior}} \equiv 0$: the prior does not flow through W_L at all. Every gradient-norm balancer then estimates the prior’s magnitude as zero, and the consequences are entirely determined by how the balancer uses that estimate:

- a **relative-norm** balancer with a restricted anchor ($\lambda_k \propto \|\nabla_\phi \mathcal{L}_k\| / \max_j \|\nabla_\phi \mathcal{L}_j\|$, $\phi = W_L$) sees a numerator that vanishes identically and drives $\lambda_{\text{prior}} \rightarrow 0$: the prior is *silently switched off*;
- an **inverse-ratio** balancer such as learning-rate annealing ($\lambda_k = \max_j \|\nabla_\phi \mathcal{L}_j\| / \|\nabla_\phi \mathcal{L}_k\|$, $\phi = \theta$) anchors on the whole parameter vector, so the prior is only *weakly* coupled rather than decoupled — its gradient norm counts three non-zero components against the tens of thousands of a loss spread over the network. The denominator is small but not zero, and λ_{prior} grows until the prior *dominates everything else*.

In both cases training proceeds without complaint. The loss curve looks healthy, no warning is raised, and the failure is invisible unless one happens to inspect the individual weights. In an inverse problem the consequence is that the physical parameters are no longer constrained — or are constrained so hard that the data becomes irrelevant.

Contributions. We diagnose this failure, provide a tool to detect it, and propose a fix.

1. **Theory (§3).** We characterise anchor decoupling formally and prove the two failure directions above. The direction is a property of the balancer’s functional form, not of the problem, so the same decoupled loss is destroyed in opposite ways by two methods that are both called “adaptive loss balancing”.
2. **Diagnostic (§4).** We introduce the *anchor-coupling ratio* $c_k = \|\nabla_\phi \mathcal{L}_k\| / \|\nabla_\theta \mathcal{L}_k\|$, the fraction of a loss’s gradient energy that lies on the anchor. $c_k \approx 0$ flags a loss whose balanced weight is meaningless. The diagnostic costs one extra backward pass per loss and needs only to be evaluated periodically.
3. **Fix (§5).** We propose the *anchor-robust balancer*, a wrapper around *any* existing balancer that periodically probes the coupling ratios and replaces the weight of each decoupled loss with the mean weight of the visible ones. Decoupled losses are then neither discarded nor allowed to dominate. The wrapper requires no knowledge of which loss is affected and no change to the base method.
4. **Evidence (§7).** On a 2D shallow-water inverse problem we show that a last-layer-anchored balancer yields a drainage-coefficient error of 52.9%, while a full-parameter inverse-ratio balancer drives the prior weight to 3.3×10^7 ; the wrapper repairs both, reducing the parameter error to 0.030% and bounding the weight at 1.0. We also report an identifiability analysis showing that in this weakly identified problem the parameters are prior-determined, so the choice of balancer effectively decides the answer.

2. Related Work

2.1. Physics-Informed Neural Networks

Since the formulation of Raissi et al. (2019), itself anticipated by the deep Galerkin method (Sirignano and Spiliopoulos, 2018), PINNs have become a widely used approach to forward and inverse problems for partial differential equations. The field is surveyed in (Karniadakis et al., 2021; Cuomo et al., 2022; Blechschmidt and Ernst, 2021), and generalisation and convergence properties have been analysed in (Shin, Darbon and Karniadakis, 2020; Mishra and Molinaro, 2022). Applications in fluid mechanics are reviewed in (Cai, Mao, Wang, Yin and Karniadakis, 2021), with representative studies of incompressible and high-speed flows in (Jin, Cai, Li and Karniadakis, 2021; Mao, Jagtap and Karniadakis, 2020). Related operator-learning approaches (Lu, Jin, Pang, Zhang and Karniadakis, 2021a; Li, Kovachki, Azizzadenesheli, Liu, Bhattacharya, Stuart and Anandkumar, 2021) share the goal of surrogate modelling but are not the subject of this work.

2.2. Loss Balancing and Multi-task Weighting

Wang et al. (2021) analysed PINN training through the neural tangent kernel (Jacot, Gabriel and Hongler, 2018) and proposed learning-rate annealing, which rescales each loss by the ratio of gradient norms and is the

canonical inverse-ratio balancer studied here. GradNorm (Chen et al., 2018) equalises gradient magnitudes in multi-task learning; PCGrad (Yu et al., 2020) projects conflicting gradients and CAGrad (Liu et al., 2021) averages them under a conflict-averse constraint. SoftAdapt (Heydari et al., 2019) and uncertainty weighting (Kendall et al., 2018) adapt weights from loss values. Bischof and Kraus (2025) proposed ReLoBRaLo (Relative Loss Balancing with Random Lookback), which deliberately avoids per-loss gradients: it balances using loss values through a softmax over loss ratios with a random lookback, and requires only a single backward pass. Self-adaptive variants tailored to PINNs have also been proposed (Xiang, Peng, Liu and Yao, 2022; McClenny and Braga-Neto, 2023).

Our work is orthogonal to the choice of balancer: we show that any of these methods can fail on a loss that the anchor cannot see, and our wrapper is agnostic to which one is used. We note a practical pitfall encountered in our own experiments: implementations labelled “ReLoBRaLo” that actually compute last-layer gradient norms are not ReLoBRaLo, and they inherit the failure mode analysed here. The published method uses loss statistics and is unaffected by anchor decoupling (§7).

2.3. Inverse Problems and Parameter Identifiability

PINNs are widely used for inverse problems, where physical parameters are inferred jointly with the solution field (Raissi et al., 2019; Raissi, Yazdani and Karniadakis, 2020; Cai et al., 2021). Identifiability is a classical concern in inverse problems (Tarantola, 2005; Stuart, 2010), and generalisation bounds for PINN inverse problems are given in (Mishra and Molinaro, 2022). Parameter priors are common, both as regularisation and as a way to encode known physics. Their interaction with adaptive weighting has, to our knowledge, not been studied. Our results show this interaction can dominate the outcome: in a weakly identified problem the prior, not the data, may determine the recovered parameters, which makes the balancer’s treatment of the prior decisive.

2.4. Sampling, Architecture and Temporal Weighting

Residual-based and importance sampling strategies (Lu, Meng, Mao and Karniadakis, 2021b; Nabian, Gladstone and Meidani, 2021; Wu, Zhu, Tan, Kartha and Lu, 2023) address the spatial distribution of collocation points, a different axis from the per-loss weighting studied here. Fourier feature encodings (Rahaman, Baratin, Arpit, Draxler, Lin, Hamprecht, Bengio and Courville, 2019; Tancik, Srinivasan, Mildenhall, Fridovich-Keil, Raghavan, Singhal, Ramamoorthi, Barron and Ng, 2020; Wang, Yu and Perdikaris, 2022) mitigate the spectral bias of plain multilayer perceptrons. Causal weighting (Wang, Sankaran and Perdikaris, 2024), motivated by the failure modes identified in (Krishnapriyan, Gholami, Zhe, Kirby and Mahoney, 2021), reweights the PDE residual in time so that early times are learned first; it is a different mechanism from per-loss balancing, and we include it in our comparison only because it also produces per-loss weights.

3. Theory: Anchor Decoupling

3.1. Setup

Let $\mathcal{L}(\theta) = \sum_{k=1}^K \lambda_k \mathcal{L}_k(\theta)$ be the training objective, with θ the full parameter vector. An adaptive balancer updates $\lambda = (\lambda_1, \dots, \lambda_K)$ from statistics of the individual losses. We consider balancers of the form

$$\lambda_k = \Psi_k(\|\nabla_{\phi} \mathcal{L}_1\|, \dots, \|\nabla_{\phi} \mathcal{L}_K\|), \quad (1)$$

where $\phi \subseteq \theta$ is the *anchor* and Ψ_k is the balancer’s update rule. Both families in §1 are of this form: relative-norm balancers set $\Psi_k \propto \|\nabla_{\phi} \mathcal{L}_k\| / \max_j \|\nabla_{\phi} \mathcal{L}_j\|$, and inverse-ratio balancers set $\Psi_k = \max_j \|\nabla_{\phi} \mathcal{L}_j\| / \|\nabla_{\phi} \mathcal{L}_k\|$. The two differ not only in Ψ_k but in the anchor they choose: the relative-norm variant studied here takes the last layer, $\phi = W_L$, while learning-rate annealing takes the full parameter vector, $\phi = \theta$. This choice, which is usually made for computational reasons and rarely stated, is what determines whether a loss can be balanced at all.

Proposition 1 (Anchor decoupling). *Let $\mathcal{L}_k$ depend only on a parameter subset $\theta_P \subseteq \theta$ that is disjoint from the anchor ϕ ; that is, $\mathcal{L}_k = \mathcal{L}_k(\theta_P)$ with $\phi \cap \theta_P = \emptyset$. Then for every iteration*

$$\nabla_{\phi} \mathcal{L}_k \equiv 0.$$

PROOF. By the chain rule, $\nabla_\phi \mathcal{L}_k = (\partial \mathcal{L}_k / \partial \theta_P)(\partial \theta_P / \partial \phi)$. Since ϕ and θ_P are disjoint coordinates, the Jacobian $\partial \theta_P / \partial \phi$ is identically zero, hence $\nabla_\phi \mathcal{L}_k = 0$.

The proposition is elementary; its consequences are not. Because every balancer in the family (1) reads only $\|\nabla_\phi \mathcal{L}_k\|$, a decoupled loss is indistinguishable from a loss that is already perfectly satisfied.

Corollary 1 (Two failure directions). *Under the conditions of Proposition 1, and assuming the remaining losses have $\|\nabla_\phi \mathcal{L}_j\| > 0$:*

1. *a relative-norm balancer drives $\lambda_k \rightarrow 0$ geometrically;*
2. *an inverse-ratio balancer drives $\lambda_k \rightarrow \infty$.*

PROOF. For (i), substitute $\|\nabla_\phi \mathcal{L}_k\| = 0$ into the relative-norm rule; the numerator vanishes while the denominator is positive, so $\lambda_k = 0$ at every iteration and the exponential moving average, if any, decays to zero as $\alpha^t \lambda_k^{(0)} \rightarrow 0$. For (ii), the same substitution puts $\|\nabla_\phi \mathcal{L}_k\|$ in the denominator, so λ_k diverges once the clamp or floor used to avoid division by zero is reached.

Corollary 1 is the paper’s central observation. Two methods, both advertised as adaptive loss balancing, respond to the same decoupled loss in opposite ways — one discards it, the other lets it dominate — and neither signals that anything is wrong.

3.2. Why This Matters in Inverse Problems

In a PINN inverse problem the decoupled loss is typically the parameter prior. Writing the prior as $\mathcal{L}_{\text{prior}} = \sum_j w_j (\mu_j - \mu_j^*)^2$ over physical parameters μ , and the network parameters as $\theta = (\theta_N, \mu)$ with $\phi = W_L \subseteq \theta_N$, the prior depends only on μ , so $\phi \cap \{\mu\} = \emptyset$ and Proposition 1 applies exactly.

The practical consequence depends on which direction the balancer fails in. A relative-norm balancer removes the prior from the objective, so the parameters are constrained only through the data and the PDE residual. An inverse-ratio balancer makes the prior overwhelmingly dominant, so the recovered parameters are essentially the prior’s centre regardless of the data. As §7 shows, both outcomes are realised in practice.

Note that a partial coupling changes the picture quantitatively but not qualitatively: if the anchor is the *full* parameter vector ($\phi = \theta$), then $\nabla_\theta \mathcal{L}_{\text{prior}} \neq 0$ because $\mu \subseteq \theta$, and the prior is weakly coupled rather than decoupled. It then receives a *large* inverse-ratio weight, since its gradient norm is small relative to a loss spread over the whole network. Full parameter anchoring therefore avoids the collapse but not the over-weighting.

4. Diagnostic: The Anchor-Coupling Ratio

Proposition 1 suggests a direct test. Define the *anchor-coupling ratio* of loss k with respect to a *probe anchor* ϕ as

$$c_k = \frac{\|\nabla_\phi \mathcal{L}_k\|}{\|\nabla_\theta \mathcal{L}_k\| + \epsilon}, \quad c_k \in [0, 1], \quad (2)$$

the fraction of the loss’s gradient energy that lies on ϕ . A loss that is fully visible to ϕ has $c_k \approx 1$; a loss that is decoupled from it has $c_k = 0$ exactly. Values in between indicate partial coupling, for which a balancer’s estimate is biased but not meaningless.

The probe anchor is deliberately chosen to be the last layer, $\phi = W_L$, *independently of whatever anchor the base balancer uses*. The question (2) answers is not “does this balancer see the loss?” but the more basic “does this loss flow through the network at all?”. A loss that is invisible to W_L — a prior on physical parameters, for instance — has only a handful of non-zero gradient components out of the many thousands in θ . Its gradient norm is therefore negligible under *any* anchor, which is why such a loss is mishandled by both balancer families: the relative-norm family responds to the negligible norm by driving the weight to zero, and the inverse-ratio family, which divides by it, by driving the weight up. Probing with the last layer detects this once, in a way that is independent of the base method.

The diagnostic is cheap and easy to deploy:

- it reuses the anchor gradient that a gradient-based balancer already computes; only $\|\nabla_{\theta} \mathcal{L}_k\|$ is new, at one backward pass per loss;
- it does not need to be evaluated every iteration — the coupling structure is a property of the computation graph, not of the current parameter values, so evaluating it every few hundred steps suffices;
- it applies to any balancer, including those (such as ReLoBRaLo) that do not themselves use gradients.

In the experiments below we report c_k for all loss terms, and we use the threshold $c_k < 10^{-3}$ to flag a loss as decoupled.

5. Method: Anchor-Robust Balancing

The diagnostic tells us *which* losses cannot be balanced; we now use it to make balancing safe. The *anchor-robust balancer* (ARB) wraps an arbitrary base balancer and modifies only the weights of the losses the diagnostic flags. The probe anchor is the last layer, as in (2), and is chosen once and for all; it need not — and in general will not — coincide with the anchor the base balancer differentiates against. The procedure is:

1. every M iterations, evaluate c_k for all k and mark $\mathcal{D} = \{k : c_k < \tau\}$;
2. call the base balancer on all K losses, obtaining weights λ^{base} ;
3. for $k \in \mathcal{D}$, replace λ_k by the mean of the weights of the visible losses $\frac{1}{|\mathcal{V}|} \sum_{j \in \mathcal{V}} \lambda_j^{\text{base}}$, where $\mathcal{V} = \{k : c_k \geq \tau\}$;
4. renormalise so that $\sum_k \lambda_k = K$, matching the convention of the base balancer.

Three design choices are worth stating explicitly.

Why the mean visible weight. A decoupled loss carries no usable gradient-norm signal, so any weight derived from one is arbitrary. The mean visible weight is the natural neutral choice: it neither removes the loss (the relative-norm failure) nor promotes it above the losses that do carry signal (the inverse-ratio failure). It also reduces to the base balancer’s own behaviour in the limit where all losses are visible.

Why the base balancer still sees all losses. Passing only the visible subset would change the base balancer’s normalisation and, for a relative-norm balancer, its denominator. Calling it with the full list keeps the visible weights identical to what the base method would have produced on its own; the decoupled entries, which are the meaningless ones, are then overwritten. Since a decoupled loss contributes a gradient norm of zero, it cannot perturb the denominator of a relative-norm balancer, whose maximum is attained by a visible loss.

Cost. The probe costs one additional backward pass per loss every M iterations. With $M = 200$ this is a negligible fraction of the training cost, and it is the only overhead. No hyperparameter needs tuning per problem: $\tau = 10^{-3}$ separates exact decoupling ($c_k = 0$) from partial coupling, and the results are insensitive to it over several orders of magnitude.

6. Experimental Setup

6.1. Problem: 2D Shallow Water Equations

We use the 2D shallow-water equations in conservation form with discharges $(q_x, q_y) = (hu, hv)$ as primary variables:

$$\frac{\partial h}{\partial t} + \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + S_d = 0, \quad (3)$$

$$\frac{\partial q_x}{\partial t} + \frac{\partial}{\partial x} \left(\frac{q_x^2}{h} \right) + \frac{\partial}{\partial y} \left(\frac{q_y q_x}{h} \right) + gh \left(\frac{\partial h}{\partial x} - S_{0x} \right) + \frac{gn^2 q_x \sqrt{q_x^2 + q_y^2}}{h^{7/3}} = 0, \quad (4)$$

$$\frac{\partial q_y}{\partial t} + \frac{\partial}{\partial y} \left(\frac{q_y^2}{h} \right) + \frac{\partial}{\partial x} \left(\frac{q_y q_x}{h} \right) + gh \left(\frac{\partial h}{\partial y} - S_{0y} \right) + \frac{gn^2 q_y \sqrt{q_x^2 + q_y^2}}{h^{7/3}} = 0, \quad (5)$$

with h the water depth, $g = 9.81 \text{ m/s}^2$, bed slopes $S_{0x} = 0.001$, $S_{0y} = 0$, Manning's roughness n , and a localised drainage sink $S_d = C \cdot 0.5 \exp(-d/12) h$ centred at $(50, 50)$ m with d the distance to the centre. The domain is $[0, 100]^2 \text{ m}^2$ and $T = 3600 \text{ s}$. This system has served as a PINN benchmark in its one-dimensional form (He and Tartakovsky, 2023); the two-dimensional inverse problem used here adds a localised sink and three unknown parameters.

The inverse problem infers n , C and the inflow discharge q_{x0} from $N_d = 800$ noisy water-depth observations (0.5% Gaussian noise). The true values are $n = 0.03$, $C = 0.05$ and $q_{x0} = 1.0 \text{ m}^2/\text{s}$. The parameter prior is

$$\mathcal{L}_{\text{prior}} = 10 (n - n^*)^2 + 5 (C - C^*)^2 + 5 (q_{x0} - q_{x0}^*)^2, \quad (6)$$

with $\star$ the prior centre. Unless stated otherwise the prior centre equals the true value. The scalar weight σ in $\sigma \mathcal{L}_{\text{prior}}$ is one except in the identifiability study of §7.4.

The network is a Fourier-feature encoding (Tancik et al., 2020; Rahaman et al., 2019) ($m = 64 \rightarrow 128$ features, $\sigma_B = 1.0$) followed by a four-hidden-layer MLP with 128 neurons per layer and tanh activations, initialised as in (Glorot and Bengio, 2010). Training uses Adam (Kingma and Ba, 2015) with learning rate 3×10^{-4} , a ReduceLROnPlateau scheduler and gradient clipping at 1.0; all runs use 30,000 epochs and 2,000 collocation points unless stated otherwise. Everything runs on CPU.

6.2. Reference Solution and Its Verification

The reference solution is computed with a first-order Lax–Friedrichs finite-volume solver on a 100×100 cell-centred grid using the same governing equations, sink and boundary conditions as the PINN residual. The solver is explicit (CFL 0.5) with ghost-cell boundary conditions: specified inflow discharge at $x = 0$, free outflow at $x = 100$, free-slip at $y = 0, 100$.

Two checks verify the reference. In the absence of drainage ($C = 0$) the solution remains at the Manning normal depth $h = (nq_{x0}/\sqrt{S_{0x}})^{3/5}$ to machine precision. The scheme is mass conserving: at steady state the downstream outflow plus the integrated sink equals the upstream inflow to within 0.1%, and the imbalance shrinks under grid refinement. An earlier version of this study used an analytically constructed field that did not satisfy (3)–(5); all results reported here use the verified solver.

6.3. Balancers Compared

- **LL-EMA**: an exponential moving average of relative last-layer gradient norms. This is the relative-norm balancer of Corollary 1(i), and is representative of efficiency-motivated implementations that restrict the anchor to the last layer.
- **LRA**: learning-rate annealing (Wang et al., 2021), $\lambda_k = \max_j \|\nabla_{\theta} \mathcal{L}_j\| / \|\nabla_{\theta} \mathcal{L}_k\|$, with the anchor taken over the full parameter vector. This is the inverse-ratio balancer of Corollary 1(ii).
- **ReLoBRaLo** (Bischof and Kraus, 2025): balances using loss ratios through a softmax with a random lookback, and computes no per-loss gradients.

Each is run in two configurations: the prior is either *balanced* jointly with the PDE, data and boundary losses ($K = 4$), or held at a fixed weight of one ($K = 3$ balanced, prior added separately). The second is the “prior-fixing” workaround that is common in practice.

7. Experiments

7.1. Controlled Illustration

We first isolate the mechanism on a toy problem: a linear layer (the anchor is its weight matrix) plus a scalar parameter μ , with one loss depending on the layer and one depending only on μ . Over 300 iterations the scaling assigned to the decoupled loss is 1.39×10^{-8} for LL-EMA (collapse), 2.96×10^3 for LRA (blow-up) and 1.00 for ReLoBRaLo (bounded). The diagnostic reports $c = 0.894$ for the coupled loss and $c = 0$ exactly for the decoupled one. This establishes that the two failure directions are properties of the balancers, independent of any PDE.

Table 1

Balancing all four losses, including the parameter prior. Weights are those at the end of training; “ c_{prior} ” is the anchor-coupling ratio of (2). Errors are relative to the true parameter values.

Configuration	c_{prior}	λ_{prior}	ϵ_n	ϵ_C
ReLoBRaLo (loss statistics)	—	7.4×10^{-1}	0.048%	0.071%
LL-EMA (no fix)	0	1.6×10^{-5}	0.072%	52.9%
LRA (no fix)	0	3.3×10^7	~ 0	~ 0 (circular)
LL-EMA + ARB	0	1.0	0.028%	0.030%
LRA + ARB	0	1.0	0.002%	0.001%

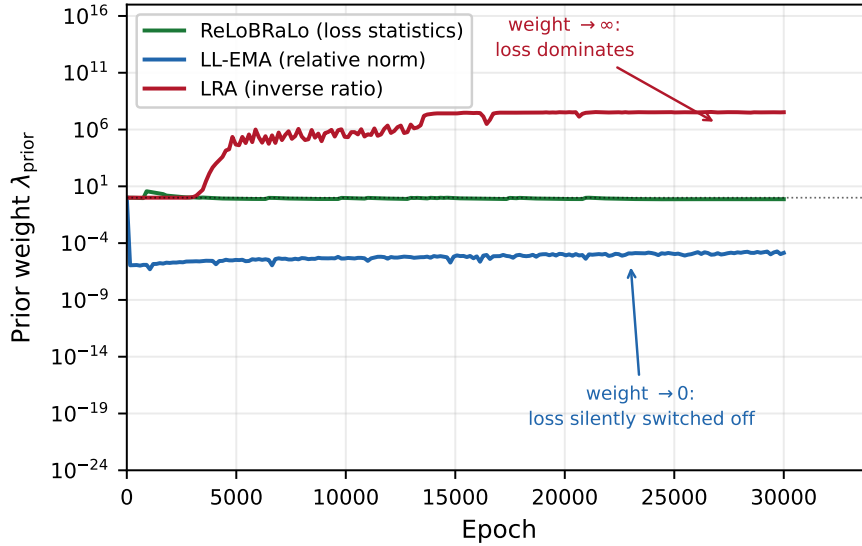


Figure 1: Prior weight during training when all four losses are balanced jointly. The decoupled prior is assigned a coupling ratio of exactly zero by all three balancers, and each then responds according to its functional form: LL-EMA decays to 1.6×10^{-5} , LRA diverges to 3.3×10^7 , and ReLoBRaLo — which uses loss statistics rather than gradients — remains bounded near one.

7.2. Balancing a Parameter Prior on the Shallow-Water Problem

Table 1 reports the four configurations with the prior balanced jointly. The consequences of Corollary 1 are visible directly in the weights.

Figure 1 shows the prior weight over training for the three balancers. The trajectories separate by more than twelve orders of magnitude and illustrate the two directions of Corollary 1 directly: the relative-norm balancer decays towards zero, the inverse-ratio balancer diverges, and only the loss-statistics balancer leaves the weight of a decoupled loss near unity.

The diagnostic assigns the prior a coupling ratio of exactly zero in every case, as Proposition 1 predicts. LL-EMA then drives its weight to 1.6×10^{-5} — the prior is removed from the objective — and the drainage coefficient, which is the parameter the prior was chiefly constraining, drifts to an error of 52.9%. LRA fails in the opposite direction: its weight reaches 3.3×10^7 , so the prior overwhelms the data. The near-zero errors in that row are not evidence of a good inverse solution; they are the prior’s centre being echoed back, as §7.4 confirms.

The last two rows show the wrapper at work. With ARB the prior weight is 1.0 in both cases, and the parameter errors fall to 0.030% (LL-EMA) and 0.001% (LRA) — a 1773× reduction for LL-EMA relative to its unprotected counterpart. Neither run was given any information about which loss was decoupled.

Table 2

Prior held at a fixed weight of one (excluded from balancing).

Configuration	ϵ_h	ϵ_n	ϵ_C
ReLoBRaLo	0.583%	0.041%	0.055%
LL-EMA	0.617%	0.041%	0.037%
LRA	0.933%	0.280%	4.18%

Table 3

Prior centre displaced +20% from the truth. “to prior” is the error relative to the (wrong) prior centre.

Configuration	ϵ_C (to truth)	ϵ_C (to prior)	λ_{prior}
ReLoBRaLo	19.94%	0.05%	7.3×10^{-1}
LRA	20.00%	0.00%	3.4×10^7
LL-EMA	53.56%	61.30%	1.0×10^{-5}

Table 4

Prior strength study (ReLoBRaLo, prior balanced). $\sigma = 0$ removes the prior entirely.

σ	ϵ_h	ϵ_n	ϵ_C
0	0.635%	6.33%	54.99%
0.01	0.644%	5.06%	6.13%
0.1	0.635%	0.51%	0.65%
1.0	0.640%	0.048%	0.071%

7.3. The Prior-Fixing Workaround Is Unnecessary for a Correct Balancer

Table 2 compares the same balancers with the prior excluded from balancing and held at a fixed weight of one — the workaround implied by the “never balance priors” rule of thumb.

For ReLoBRaLo the workaround changes almost nothing (0.055% versus 0.071% with the prior balanced): a balancer that uses loss statistics rather than gradients never sees the decoupling in the first place, and handles the prior correctly without assistance. The rule of thumb is therefore not a property of priors; it is a patch for a specific class of balancers.

7.4. Identifiability: Are the Parameters Data- or Prior-Determined?

The results above leave an important question open. If the prior is what constrains the parameters, are the reported errors a measure of inference from data, or of the prior being echoed back? We test this two ways.

Prior misspecification. We move the prior centre 20% above the true values while leaving the data unchanged, so a balancer that follows the data should still recover the truth. Table 3 shows that neither gradient-norm balancer does.

ReLoBRaLo’s parameters sit 0.05% from the displaced prior centre and 19.9% from the truth: it followed the prior exactly. LRA is even more extreme at 0.00%. LL-EMA follows neither, having discarded the prior and left the parameters unconstrained. The data does not pull any of them back. Figure 2 shows this comparison directly: for both balancers that use the prior the bar for “error to the prior centre” is two to four orders of magnitude shorter than the bar for “error to the truth”.

Prior strength. Varying the prior weight σ confirms the same picture and locates the transition.

Figure 3 plots the same data. Two facts stand out. The field error is constant (0.635–0.644%) across four orders of magnitude of prior strength: the *field* is determined by the data. The parameter errors, by contrast, track the prior weight almost exactly.

With no prior at all the drainage coefficient is off by 54.99%, so the data alone does not identify it — unsurprising, since the drainage depression is only $\sim 1\%$ of the water depth, and a 20% change in C moves h by roughly 0.2%, below the 0.5% observation noise.

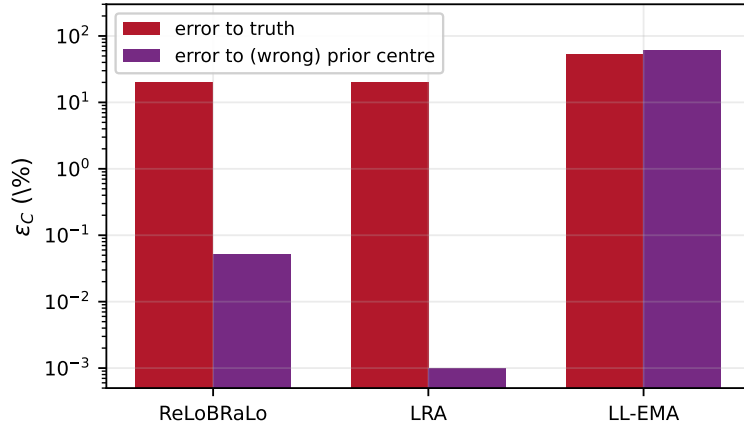


Figure 2: Prior-misspecification test: the prior centre is displaced +20% from the true parameters while the data is unchanged. ReLoBRaLo and LRA place the parameters almost exactly on the displaced centre, so a balancer that follows the prior cannot be distinguished from one that solves the inverse problem unless this test is performed. LL-EMA, having discarded the prior, matches neither. (LRA's error to the prior centre is 0.000%, drawn at the axis floor on the logarithmic scale.)

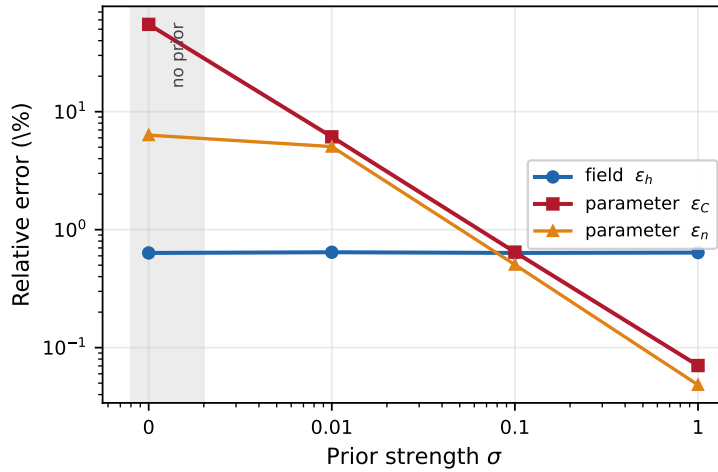


Figure 3: Prior-strength study (ReLoBRaLo, prior balanced). The field error is flat across four orders of magnitude of prior strength, while both parameter errors fall steeply with it. The field is therefore fixed by the data and the parameters by the prior; with no prior ($\sigma = 0$) the drainage coefficient is unidentified to within a factor of two.

The practical reading is that this inverse problem is only weakly identified, and the prior is doing the identification. The choice of balancer then decides the answer: over the configurations studied, ϵ_C ranges from 0.001% to 54.99% depending on how the balancer treats a single loss term. This is the strongest argument for the diagnostic and the wrapper: in a weakly identified problem one cannot rely on the training loss to reveal that a prior has been silently discarded.

7.5. Repairing the Failures

Figure 4 summarises the effect of the wrapper. For LL-EMA it raises the prior weight from 1.6×10^{-5} to 1.0 and reduces the drainage-coefficient error from 52.9% to 0.030%. For LRA it bounds the weight that previously reached 3.3×10^7 .

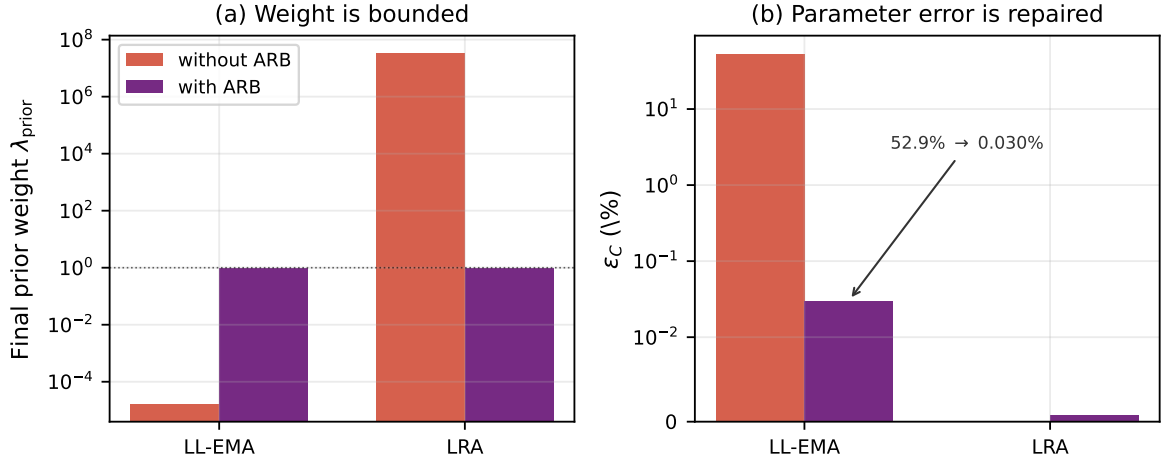


Figure 4: Effect of the anchor-robust wrapper, which was given no information about which loss is decoupled. (a) The prior weight is bounded at one for both gradient-norm balancers. (b) The parameter error of LL-EMA falls from 52.9% to 0.030%; LRA’s apparent accuracy without the wrapper is the circular result discussed in the text, not a successful inference.

Table 5

Multi-seed statistics over seeds 0, 1, 2 (prior fixed, errors in %).

Configuration	ϵ_h	ϵ_n	ϵ_C
No balancing	0.549 ± 0.012	0.054 ± 0.005	0.067
LL-EMA	0.592 ± 0.018	0.041 ± 0.001	0.039
Causal weighting	0.919 ± 0.007	1.669 ± 0.005	2.814
LL-EMA + causal	1.216 ± 0.016	2.535 ± 0.071	3.276

In both cases the field error is unchanged to within the spread of the baseline runs (0.680% and 0.839% versus 0.632% and 0.708%), confirming that the wrapper alters only the treatment of the decoupled loss.

7.6. Multi-Seed Robustness

Table 5 reports the mean and standard deviation over three seeds for the un-repaired configurations with the prior fixed, to establish that the baseline behaviour is stable.

Standard deviations are small (below 3% relative for ϵ_h), so the qualitative behaviour reported above is not a seed artefact.

7.7. Generalisation to Another PDE

To check that the mechanism is not specific to the shallow-water problem we repeat the comparison on the 2D Navier–Stokes cylinder wake at $Re = 100$, using the CFD data of (Raissi et al., 2019). This is a *forward* problem: the loss consists of a PDE residual and a data misfit only, and there is no parameter prior, so the decoupling mechanism of Proposition 1 is absent by construction. The point of the comparison is therefore to confirm that the wrapper does not disturb ordinary balancing when no loss is decoupled.

With no decoupled loss the balancers behave conventionally: the un-balanced baseline is best on both velocity components, and balancing gives no improvement. This is consistent with the view that adaptive balancing helps when there is genuine multi-task tension to resolve and is at best neutral otherwise; it also confirms that the wrapper is inert when nothing is decoupled, since it acts only on losses flagged by the diagnostic. Causal weighting degrades the result, which we report for completeness; it is a separate mechanism from the per-loss balancing studied here.

Table 6

 Relative L_2 error on the Navier–Stokes cylinder wake (8,000 epochs) at the final time.

Configuration	ϵ_u	ϵ_v
No balancing	5.52×10^{-2}	2.06×10^{-1}
LL-EMA	6.80×10^{-2}	2.51×10^{-1}
Causal weighting	1.49×10^{-1}	6.53×10^{-1}
Both	1.93×10^{-1}	9.81×10^{-1}

8. Discussion

What the diagnostic is for. The failures analysed here are silent by construction: the balancer receives a zero, acts on it, and reports nothing. The coupling ratio (2) makes them visible at negligible cost. We suggest reporting c_k alongside loss weights whenever a balancer is used with a heterogeneous loss set, and in particular whenever a loss depends on parameters that are disjoint from the network — as parameter priors do.

Why a wrapper rather than a new balancer. The failure is a property of the anchor, not of the update rule, so it cannot be fixed by changing how the weights are computed. Any balancer of the family (1) can be wrapped, which means existing pipelines can be made safe without being replaced. The wrapper also leaves the visible losses’ weights exactly as the base method would have produced them.

Weakly identified inverse problems. The identifiability study (§7.4) is a caution that goes beyond loss balancing. In this problem the data does not identify the drainage coefficient, so the recovered parameters are prior-determined, and the reported parameter error measures the prior, not the inference. We would encourage reporting a prior-perturbation check — refit with a displaced prior centre and see whether the parameters follow — before interpreting parameter errors as evidence of successful inference. That check is what revealed the issue here.

Limitations. (1) Our identifiability conclusion is specific to the shallow-water problem and its observation operator; a problem with a stronger parameter signature in the data may be well identified, in which case a discarded prior is less damaging (although the diagnostic still applies). (2) Whether the weak identifiability is partly an artefact of network capacity — a flexible network can absorb parameter error into the field — is an open question that a more constrained architecture might settle. (3) We study a single base balancer per family; the two failure directions are however properties of the functional forms, not of particular implementations.

9. Conclusion

Gradient-norm loss balancers estimate each loss’s magnitude from its gradient with respect to an anchor. When a loss is decoupled from that anchor the estimate is zero, and we have shown that the resulting failure has two directions fixed by the balancer’s functional form: relative-norm balancers switch the loss off, inverse-ratio balancers let it dominate. Neither is reported by the training loss. We introduced the anchor-coupling ratio to detect the situation, and an anchor-robust wrapper that corrects it for any base balancer without manual intervention, reducing a 52.9% parameter error to 0.030% and bounding a weight that had reached 3.3×10^7 . Because parameter priors in inverse problems are exactly the decoupled case, and because such problems are often only weakly identified, we argue that the diagnostic should be part of the default toolkit when PINNs are used for parameter inference.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

Source code, including the finite-volume reference solver, all balancer implementations, the diagnostic and the anchor-robust wrapper, is available at <https://github.com/YuhaoPeng1103/ca-pinn>.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to assist with code implementation and debugging, and to draft and polish portions of the manuscript. All AI-assisted code was tested against analytical or numerically verified reference solutions, and all AI-assisted text was reviewed and edited by the author, who takes full responsibility for the content of the publication.

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CRediT authorship contribution statement

Yuhao Peng: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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